

Tier 1: Macro Plateau Controller π_{macro}

- Semi-MDP over segments $\Delta_{\text{seg}} = 100$ iterations
- Dueling DDQN with Prioritized Replay (Q_{macro})
- 13-dim state $\mathbf{s}_k^{\text{macro}} \rightarrow 7$ strategic regimes (a_0 – a_6)

Trigger a_5

Mode a_k (every Δ_{seg} iters)

Tier 2: Micro Selection Tier (π_{micro} & LAC)

- Dueling DDQN over 20-dim state vector $\mathbf{s}_t^{\text{micro}}$
- Softmax-Entropy concentration gate $w_{\text{conf}}(\mathbf{s})$
- Blends neural Q-values & bandit priors (65 pairs)
- Learned Acceptance (LAC) lookahead ($H_{\text{lac}} = 80$)

Operator pair (d, r) (iter t)

Tier 3: Heuristic Execution Engine (ALNS)

- 13 Destroy $\times$ 5 Repair composite operators
- $O(1)$ forward-time-slack feasibility evaluation
- Top- k ($k = 25$) candidate pruning + GNN filter
- Polynomial Generalized Ejection Chains (GEC)

Feasible s_{cand}

Dual-Slot Route Pool $\mathcal{P}$

- Capacity $|\mathcal{P}|_{\text{max}} \leq 2000$ (frozenset dedup)
- Slot A (75%): Cost-per-stop metric $c_r/|r|$
- Slot B (25%): Sub-routes for fleet reduction

Columns in $\mathcal{P}$

Set Partitioning Recombination (HiGHS)

- HiGHS MIP solve over route pool $\mathcal{P}$ (4.0s cutoff)
- Geometric penalty schedule $\lambda_{\text{pen}} \in \{2 \dots 50\}\bar{c}$
- Certified penalty guarantees exact $N_V \succ D_T$

Recombined s_{MIP}

Incumbent s_{inc} & Elite Archive

- Feasibility invariant (100% TW & capacity)
- Lexicographic admission criterion ($N_V \succ D_T$)
- Independent cold-start (zero inter-run leakage)

Elite routes $\rightarrow \mathcal{P}$

$\mathbf{s}_k^{\text{macro}} \in \mathbb{R}^{13}$

$\mathbf{s}_t^{\text{micro}} \in \mathbb{R}^{20}$